

* Count Factor -:

$N = 24 \rightarrow 1, 2, 3, 4, 6, 8, 12, 24$

$\rightarrow 8$ factors

$i * j = N \rightarrow i \text{ \& } j \text{ are factors of } N$

$$\Rightarrow j = N/i$$

$\rightarrow i \text{ \& } N/i \text{ factors of } N$

* Prime check -:

That exactly two factors

$\Rightarrow$ Neither prime nor composite

* Sum 1-to-::

Sum of first N Natural Numbers

$$S(N) = \frac{N(N+1)}{2}$$

* Square root -:

Example $N = 25 \rightarrow 5$

* Logarithm :-

Base $\rightarrow$ $2^3 = 8 \rightarrow$ value

$3 \rightarrow$ power / Exponent

Eq. $\log_2 64 = 6$

$$\log_a^N = N$$

$$2^k = N$$
$$\log_2 N = k$$

Q Given a positive integer N , how many times do we need to divide it by 2 until it reduces to 1

$$N = 16$$

$$16 \rightarrow 8 \rightarrow 4 \rightarrow 2 \rightarrow 1 \quad 4 \text{ times}$$

$$\log_2 N = k$$

